

Foundations of Data Science for Biologists

Data wrangling: merges and joins

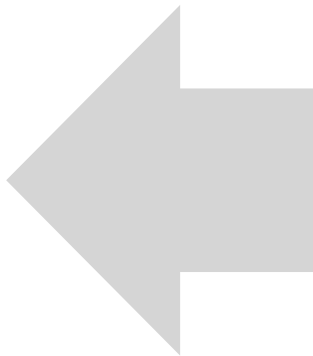
BIO 724D

2026-09-22

Instructors: Paul Magwene, Greg Wray

Merge operations

Date	Location	MinTemp	MaxTemp
2009-01-01	WaggaWagga	9.9	28.4
2009-01-02	WaggaWagga	8.9	25.3
2009-01-03	WaggaWagga	11	30.7
2009-01-04	WaggaWagga	14.7	35.2
2009-01-05	WaggaWagga	11.8	36.5
2009-01-06	WaggaWagga	12.5	39.1
2009-01-07	WaggaWagga	15.6	41.8
2009-01-08	WaggaWagga	14.5	30.4



Date	Location	Rainfall	Evaporation
2009-01-01	WaggaWagga	0	9.8
2009-01-02	WaggaWagga	0	14.8
2009-01-03	WaggaWagga	0	9.8
2009-01-04	WaggaWagga	0	9.4
2009-01-05	WaggaWagga	0	11.6
2009-01-06	WaggaWagga	0	13
2009-01-07	WaggaWagga	0	15.2
2009-01-08	WaggaWagga	0	17



Date	Location	MinTemp	MaxTemp
2009-01-09	WaggaWagga	14.8	28.7
2009-01-10	WaggaWagga	15.9	29.7
2009-01-11	WaggaWagga	18.1	31.3
2009-01-12	WaggaWagga	16.5	35.4

Merge operations

Date	Location	MinTemp	MaxTemp	Date...5	Location...6	Rainfall	Evaporation
2009-01-01	WaggaWagga	9.9	28.4	2009-01-01	WaggaWagga	0	9.8
2009-01-02	WaggaWagga	8.9	25.3	2009-01-02	WaggaWagga	0	14.8
2009-01-03	WaggaWagga	11	30.7	2009-01-03	WaggaWagga	0	9.8
2009-01-04	WaggaWagga	14.7	35.2	2009-01-04	WaggaWagga	0	9.4
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2009-01-06	WaggaWagga	12.5	39.1	2009-01-06	WaggaWagga	0	13
2009-01-07	WaggaWagga	15.6	41.8	2009-01-07	WaggaWagga	0	15.2
2009-01-08	WaggaWagga	14.5	30.4	2009-01-08	WaggaWagga	0	17
2009-01-09	WaggaWagga	14.8	28.7	2009-01-09	WaggaWagga	0	14
2009-01-10	WaggaWagga	15.9	29.7	2009-01-10	WaggaWagga	0	11.4
2009-01-11	WaggaWagga	18.1	31.3	2009-01-11	WaggaWagga	0	10.4
2009-01-12	WaggaWagga	16.5	35.4	2009-01-12	WaggaWagga	0	7.2

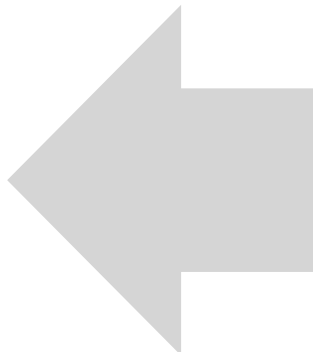
Removes redundant header row (if columns match)

All columns are retained, duplicating information (changes names if duplicated)

Not content-aware: if rows are mis-aligned, data will be scrambled

Join operations

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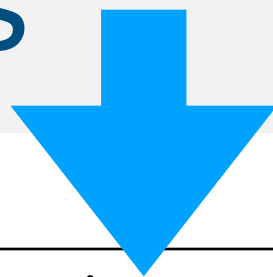


Location	Province	Geolocation	Elevation
CoffsHarbor	NSW	-30.298504° 153.114330°	7
WaggaWagga	NSW	-35.104832° 147.364811°	190
Tuggeranong	ACT	-35.419780° 149.068679°	580
Brisbane	VIC	-37.565288° 143.844779°	450
Mildura	VIC	-34.201612° 142.136100°	45
Cairns	QLD	-16.925742° 145.770949°	6
MountGambier	SA	-37.830081° 140.779637°	55
SalmonGums	WA	-32.982586° 121.643815°	257
AliceSprings	NT	-23.699023° 133.880747°	582
Darwin	NT	-12.461995° 130.841583°	27
Uluru	NT	-25.343378° 131.042680°	852
Launceston	TAS	-41.439895° 147.134793°	11

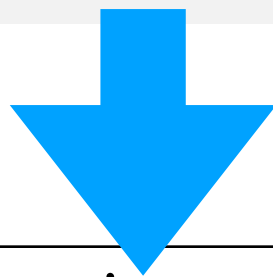
Define one or more join columns

When values match, join rows

Join operations



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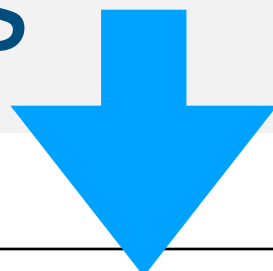


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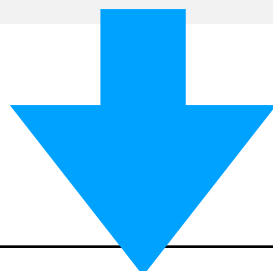
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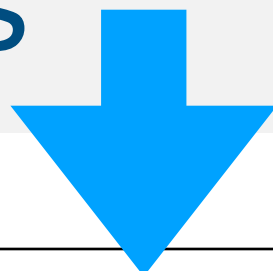


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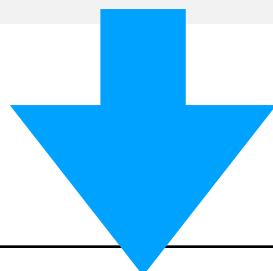
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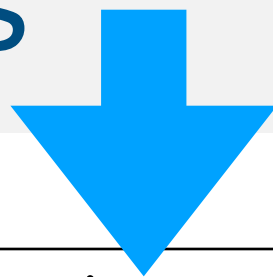


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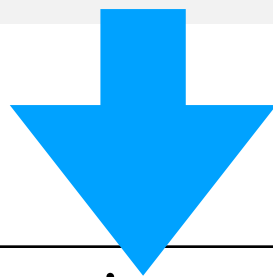
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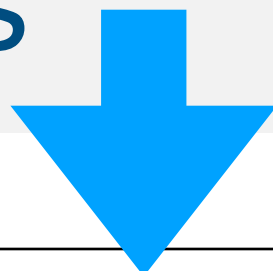


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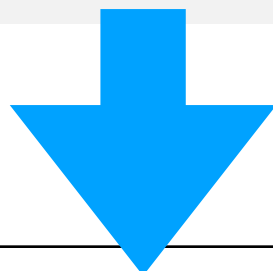
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When values match, join rows

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2008-12-04	Ballarat	6.7	21.8	NA	NA		NA
2008-12-05	Ballarat	13.3	25.6	NA	NA		NA
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2009-01-02	Mildura	8.9	24.8	VIC	-34.201612°	142.136100°	45
2009-01-03	Mildura	10.6	28.6	VIC	-34.201612°	142.136100°	45
2009-01-04	Mildura	13.2	34.5	VIC	-34.201612°	142.136100°	45
2009-01-05	Mildura	16.5	37.3	VIC	-34.201612°	142.136100°	45

Less duplication

Position-independent

Value-based

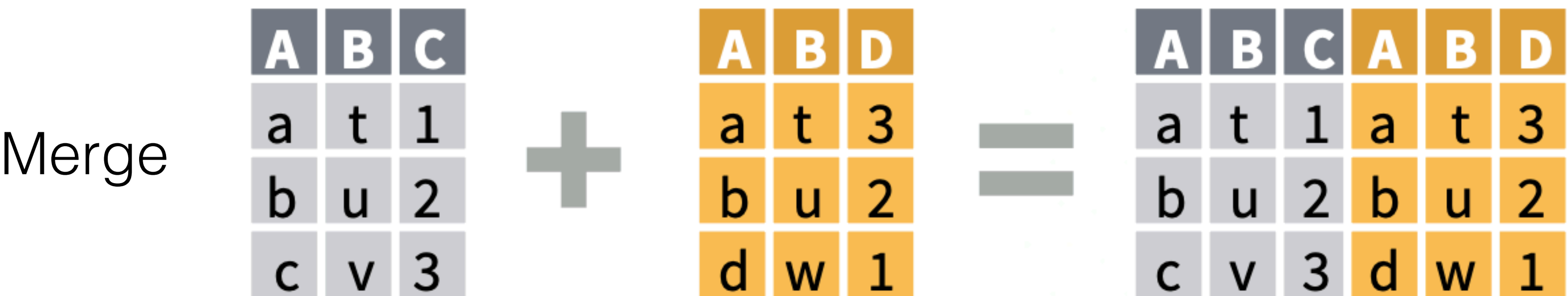
NA if no match

Highly flexible

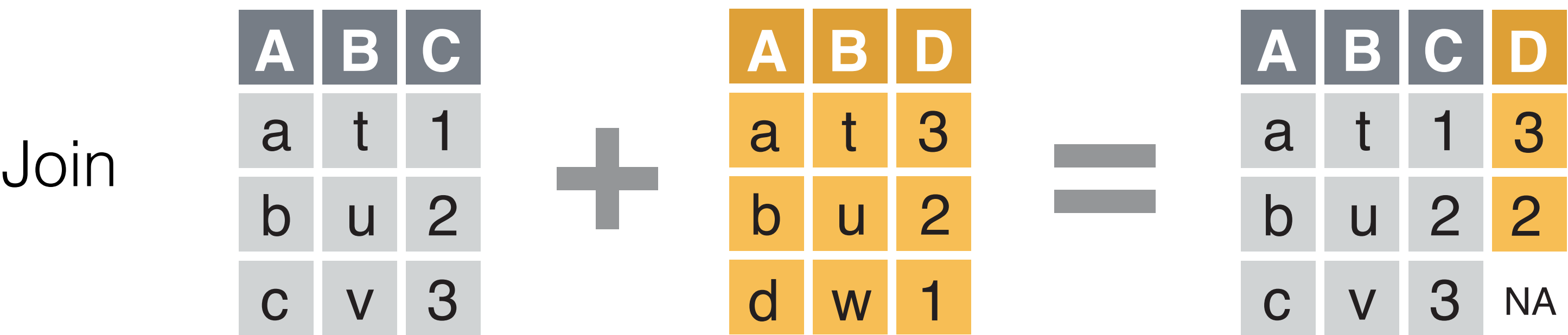
What are merge and join operations?

Both are useful and common ways to combine information from two or more data frames

They produce different outcomes and have different use cases



Appends data
Requires same structure
Brittle



Integrates data
Allows different structure
Robust

Data frame merges

Merge operations

Merge rows

The diagram illustrates the addition of two matrices, x and y , to produce a resulting matrix z .

Matrix x :

	A	B	C
a	t	1	
b	u	2	
c	v	3	

Matrix y :

	A	B	C
c	v	3	
d	w	4	

Resulting Matrix z :

	A	B	C
a	t	1	
b	u	2	
c	v	3	
c	v	3	
d	w	4	

Add more observations (long-form)

Merge columns

Diagram illustrating the addition of two matrices:

A	B	C
a	t	1
b	u	2
c	v	3

+

A	B	D
a	t	3
b	u	2
d	w	1

=

A	B	C	A	B	D
a	t	1	a	t	3
b	u	2	b	u	2
c	v	3	d	w	1

Add more observations (wide-form)

Add more attributes (long-form)

Considerations when merging **rows**

Check that the two data frames are compatible before you try to merge

- The two data frames should have the same columns (meaning, not just name)

- Columns are matched by name, not by position

- Any missing column will be filled with **NA** values

- Check for the expected number of columns after the merge using **length()**

Decisions about how to merge:

- Which set logic to use (see next slide)

- Whether you want to retain information about table of origin (see documentation)

Warning: **dplyr** merge functions are quite forgiving

- It is up to you to ensure that the merge makes sense!

Considerations when merging **columns**

Check that the two data frames are compatible before you try to merge

- Check for equivalent number of rows using the function `nrow()`

- If adding additional observations, make sure rows align correctly

- If adding additional attributes, makes sure values match the correct rows

During the merge, column names are checked and repaired by default

- Duplicates are re-named and names are added if not present

All data are retained without modification or filtering

- Columns with the same name and columns with identical values are all kept

Warning: `dplyr` merge functions are quite forgiving

- It is up to you to make certain that the merge makes sense!

Data frame joins

What is a join? How does it differ from a merge?

Join: combine data in two tables based on values in one (or more) columns

Row order doesn't matter!

Values in the join column(s) dictate how data are combined

Best way to understand is with diagrams on following slides

Process of combining data differs:

Merge: attach rows or columns in order, ignoring values

Join: attach data based on values in the join column(s), ignoring order

Outcome thus differs:

Merge: all data are preserved; row and column order is preserved

Join: some data may be removed; row and column order may be rearranged

Specifying a join

Joins rely on matching values in one or more **columns** present in both data frames

Typically, just one column is matched (e.g., a unique ID)

However, matching in multiple columns is possible (e.g., genus and species)

Decisions about how to join

Set logic determines which rows and columns are retained and how they are attached

Duplicate rows can be retained or discarded

A	B	C		A	B	D
a	t	1	+	a	t	3
b	u	2		b	u	2
c	v	3		d	w	1

The following examples join on column **A** only

This is different from default dplyr behavior

Left join

Left joins (also known as left outer joins) are by far the most common type of join

x				y				result			
A	B	C	+	A	B	D	=	A	B	C	D
a	t	1		a	t	3		a	t	1	3
b	u	2		b	u	2		b	u	2	2
c	v	3		d	w	1		c	v	3	NA

Left joins create a union of matches to the first (left) data frame: order matters!

- Keep all rows from x

- Discard rows in y that do not match the value in column A of x

- Discard columns from y that duplicate those in x

- Attach columns from y where there is a match in column A; attach NAs if no match

Inner join

Inner joins are another type of join operation

x				y				result			
A	B	C		A	B	D		A	B	C	D
a	t	1	+	a	t	3	=	a	t	1	3
b	u	2		b	u	2		b	u	2	2
c	v	3		d	w	1					

Inner joins create an intersection of matches between data frames: order does not matter

- Keep rows in x that match y in column A

- Discard rows from x that do not match y in column A

- Discard columns from y that duplicate those in x

- Attach columns from y where there is a match in column A

Full join

Full joins are rare, but useful in some specific situations

x				y				result			
A	B	C		A	B	D		A	B	C	D
a	t	1	+	a	t	3	=	a	t	1	3
b	u	2		b	u	2		b	u	2	2
c	v	3		d	w	1		c	v	3	NA
								d	w	NA	1

Full joins create a union of all rows and columns

- Keep all rows in x

- Add all rows from y that do not match x

- Attach columns from y when a match

- Insert NA where rows from y lack a column from x, and vice-versa

Comparing join operations

Source tables

X			y		
A	B	C	A	B	D
a	t	1	a	t	3
b	u	2	b	u	2
c	v	3	d	w	1

Join methods

A	B	C	D
a	t	1	3
b	u	2	2
c	v	3	NA

Left join

Union of matches to left (table x)

A	B	C	D
a	t	1	3
b	u	2	2

Inner join

Intersection of matches

A	B	C	D
a	t	1	3
b	u	2	2
c	v	3	NA
d	w	NA	1

Full join

Union of all rows and columns

